nftables workshop

NetDev 1.2 Tokyo, Japan

Pablo Neira Ayuso
<pablo@netfilter.org>

nf updates since last netdev

- Per-netns hooks: xtables registers hooks on demand
- Xtables input sanitization for unpriviledged netns
- Single hashtable for all namespaces
- align nf_conn on cacheline boundary
- convert nat hash to use rhashtable
- move frequently dereferenced fields into struct nf_conn
- remove ip_conntrack* sysctl compat code
- get rid of conntrack timers
- Disable automatic helper assignment
- Hashlimit 64-bits pps resolution

nftables updates since last netdev

- Native nfqueue for bridge family
  - Another step forward to deprecate br_netfilter
- Set infrastructure updates:
  - Allow adjacent ranges in set representation
  - Introduce add/create semantics
  - Support deletion of inactive elements
  - New flush set elements command
  - Inversion for lookups and dynamic element insertion from packet path
- New expressions: Numgen, hash, range, quota.
- Connlabel support
- Nfqueue integrate with maps
- Many fixes...

Revisiting nf hooks

- Conversion from double-linked list → Single-linked list (4.9)
  - Next step: Conversion to array of hooks (scheduled for 4.10)
- Deprecate NF_STOP
  - Equivalent to NF_ACCEPT
- Move NF_QUEUE handling away from core
  - Run nf_queue() from xt_NFQUEUE
- Only 3 verdicts to handle after this
- nf_hook_slow() becomes small
  - Inline it.

Other fronts

- Switch offload:
  - Marvell 88e6xxx
    - 255 rule entries in tcam, rule based, match + actions
  - Qualcomm QCA8337
    - flow table 2048 entries, updated from packet path
- NPU offload
- jit via bpf PoC

Programmability: nft way



- We have a network specific purpose VM
  - Very simple bytecode verification
  - 22 instructions
  - 32/128 addressable regs
  - Provides netlink interface
- Stack datapath is unflexible so...
- Let's reuse existing components in the kernel
  - By adding new nft instructions that integrate with them
  - Use existing netlink interface to configure them
- Then, plug nft instructions in lego fashion to define your datapath.
- Use rule-based syntax combined with maps
  - No linear rule inspection anymore!